Can superparamagnetic iron-oxide nanoparticles be seen in CT? A CONRAD-native simulation study

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ABSTRACT

Superparamagnetic iron-oxide nanoparticles (SPIONs) are used for magnetic drug targeting, hyperthermia, and cell tracking. A simple question is often left open: can the deposited iron be seen in a computed tomography (CT) scan? We answer this by simulation. We start from a real magnetite SPION formulation. We build a delivered-mass dose model. A fixed particle mass is spread through an $8\,\mathrm{cm}^3$ tumor. At realistic loading this leaves only a fraction of a milligram of iron per millilitre. That is about ten times below iodine CT enhancement. Iron has no usable K-edge in the diagnostic range. Its contrast is photoelectric and lives at low energy. We simulate the whole particle: a magnetite core plus a polyacrylic-acid coating, with the coating fraction taken per formulation from the article's thermogravimetry. The imaging chain is CONRAD-native. It uses a fan-beam short scan with Parker weighting, cosine and Shepp-Logan filtering, a distance-weighted backprojector, and an always-on water beam-hardening precorrection. We compare an energy-integrating detector (EID) with a photon-counting detector (PCD). We optimize the tube voltage, the aluminium filtration, and the PCD energy bins for iron. We score detectability by iron ΔHU, contrast-to-noise ratio (CNR), and the Rose criterion. We study homogeneous cellular uptake and vascular delivery, and verify the result in three dimensions on rabbit anatomy. The optimum is a soft 70 kVp beam with 5 mm aluminium, which gives an iron CNR of 16.9 at 1 mg Fe/ml, about 1.2× the energy-integrating detector. The lowest detectable iron at the Rose criterion is about 0.4 mg Fe/ml for the photon-counting detector at the optimum. The finding is that SPIONs sit near the CT detection limit at realistic load, and that tumor cell density is the dominant factor for visibility.

Keywords: SPION, iron oxide nanoparticles, computed tomography, photon counting, detectability, contrast-to-noise ratio, spectral optimization, simulation, CONRAD

1. INTRODUCTION

Superparamagnetic iron-oxide nanoparticles (SPIONs) are a workhorse of nanomedicine. Magnetic fields steer them for local drug delivery. They can be heated for hyperthermia. They can label cells for tracking. The particles we study are those of Heinen et al.[2] They are polyacrylic-acid (PAA)-coated magnetite (Fe_3O_4) clusters. The cores are about 8 and 12 nm. They assemble into hydrodynamic clusters of about 70–250 nm. They are used at iron concentrations of about 1–10 mg Fe/ml.

A clinical question follows. Once these particles reach a target, can we simply see them in a routine CT scan? The answer matters. It matters for dose planning. It matters for reading incidental findings. It matters most for image-guided delivery, where one wants to confirm that particles reached the target without a second imaging modality.

At first sight iron looks promising. It is denser and higher in atomic number than soft tissue. But the amount is the problem. At biologically realistic loading the delivered iron mass is small. A few milligrams of particles spread through a cubic-centimetre tumor leave only tenths of a milligram of iron per millilitre. That is roughly ten times below the iodine concentrations that give comfortable CT enhancement. Whether this little iron rises above image noise is not obvious.

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The physics is also unhelpful. Iron has a K-edge at 7.1 keV. This is far below the diagnostic window. So there is no K-edge jump to exploit, unlike iodine or gadolinium. The iron contrast is purely photoelectric. It is strongest at low energy and fades quickly as energy rises. This shapes every result that follows: a soft spectrum helps, a hard spectrum hurts, and photon-counting can only help where low-energy photons survive the body.

Despite a large SPION literature and a surge of interest in photon-counting CT, there is little quantitative simulation of SPION CT detectability across modern detectors. We address this gap. Our contributions are: (i) a delivered-mass dose model tied to a real magnetite SPION formulation, with a whole-particle (core + coating) material model and a per-formulation coating fraction from thermogravimetry; (ii) a CONRAD-native fan-beam pipeline in which every reconstruction step is a CONRAD class; (iii) a spectrum-and-detector optimization that picks the CNR-optimal tube voltage, aluminium filter, and PCD energy bins for iron; (iv) two biological uptake models – homogeneous cellular uptake and vascular fresh injection – grounded in the article's measured cellular loading; and (v) a three-dimensional cone-beam verification on real rabbit anatomy. The pipeline is open and every intermediate result is public for audit.

2. MATERIALS AND METHODS

We fix the parts in order. First the nanoparticle and dose model. Then the phantom. Then the CONRAD-native imaging chain. Then the two detectors. Then the experiments and metrics. Each choice keeps the $c_{\mathrm{Fe}} = 0$ case identical to background, so any measured contrast is attributable to iron alone.

2.1 Nanoparticle and delivered-mass dose model

We model the whole particle, not iron alone. The whole particle is a magnetite (Fe_3O_4) core plus a PAA coating (monomer $C_3H_4O_2$). The iron mass fraction of magnetite is 0.724 ($3 \times 55.85/231.5$). The bound oxygen is nearly tissue-equivalent. It adds only a few percent to the attenuation per gram of iron. The PAA coating is low-Z (C/H/O). It is essentially water-equivalent. It is therefore negligible for the X-ray attenuation, but we still include its mass so the reported particle concentration is honest.

We keep two concentrations distinct throughout. The iron concentration c_{Fe} (mg Fe/ml) drives the attenuation. The whole-particle concentration c_{NP} (mg SPION/ml) is derived from it. The link is the coating fraction. From an iron concentration we recover the whole-particle concentration by

$$c_{\mathrm{NP}} = \frac{c_{\mathrm{Fe}}}{0.724\,(1-\varphi)}, \tag{1}$$

where φ is the PAA coating mass fraction. All contrast numbers are reported as mg Fe/ml in the tumor.

The coating fraction is taken per formulation from the article's supplementary thermogravimetry (TGA).[2] The inorganic (iron-oxide) residual is 87.7% for SPION I (12 nm cores) and 66.4% for SPION II (8 nm cores). On heating the residual oxidizes magnetite to hematite, adding 3.4% mass. Correcting for this, the original magnetite fraction is 84.8% for SPION I and 64.2% for SPION II. The complement is PAA. So $\varphi \approx 0.15$ for SPION I and $\varphi \approx 0.36$ for SPION II. SPION II carries roughly twice the coating. Each formulation is therefore simulated with its own coating: $c_{\mathrm{NP}} \approx 1.63\,c_{\mathrm{Fe}}$ for SPION I and $c_{\mathrm{NP}} \approx 2.16\,c_{\mathrm{Fe}}$ for SPION II. Adding the coating mass moves the monochromatic iron ΔHU by only about 3.6% at $\varphi = 0.15$, so the attenuation stays iron-dominated. Table 1 lists the composition of the two formulations.

We use a delivered-mass dose model, not a fixed vial concentration. We spread a fixed particle mass through the tumor and ask how much iron that leaves per millilitre. We anchor at 6 mg of SPIONs delivered for the 10 mg/ml formulation over an 8 cm^3 tumor and scale with concentration. This gives the linear relation

$$c_{\mathrm{Fe}} = 0.0543 \cdot c_{\mathrm{form}} \quad [\mathrm{mg\,Fe/ml}], \tag{2}$$

where c_{form} is the formulation concentration in mg/ml. The reference 6 mg dose gives $c_{\mathrm{Fe}} = 0.54$ mg Fe/ml. Even the top level of the sweep is only about 1 mg Fe/ml. This is by design: we model the diluted delivered dose in the tumor, not the concentrated vial.

The material attenuation is taken from CONRAD's database. As a cross-check we compared the magnetite mass attenuation against an independent NIST-based model. The two agree to five decimals, so the physics underlying every result is validated. At a representative 60 keV, 1 mg Fe/ml raises the attenuation by only about 6.7 HU, which sets the scale of the whole study. The iron K-edge sits at 7.1 keV, far below the diagnostic range, so there is no edge to exploit. Figure 1 shows the linear attenuation of soft tissue, cortical bone, and iron-loaded tumor, and where the iron contrast lives in energy.

Table 1. Nanoparticle composition (E1). The whole particle is a magnetite core plus a PAA coating. The coating fraction φ is from the article's TGA. Iron mass fraction of magnetite is 0.724. c_{NP} per unit c_{Fe} follows Eq. (1).

Component	Description	Value
Core	Magnetite Fe_3O_4	iron fraction $F_{\mathrm{Fe}} = 0.724$
Coating	Polyacrylic acid (PAA), $(C_3H_4O_2)_n$	low-Z, tissue-equivalent
φ (SPION I)	PAA mass fraction (TGA, 12 nm core)	0.15
φ (SPION II)	PAA mass fraction (TGA, 8 nm core)	0.36
Example conversion $c_{\mathrm{Fe}} = 1.0\,\mathrm{mg\,Fe/ml} \rightarrow$ whole-particle c_{NP}:		
SPION I	$c_{\mathrm{NP}} = c_{\mathrm{Fe}}/[F_{\mathrm{Fe}}(1-\varphi)]$	1.62 mg SPION/ml
SPION II		2.16 mg SPION/ml

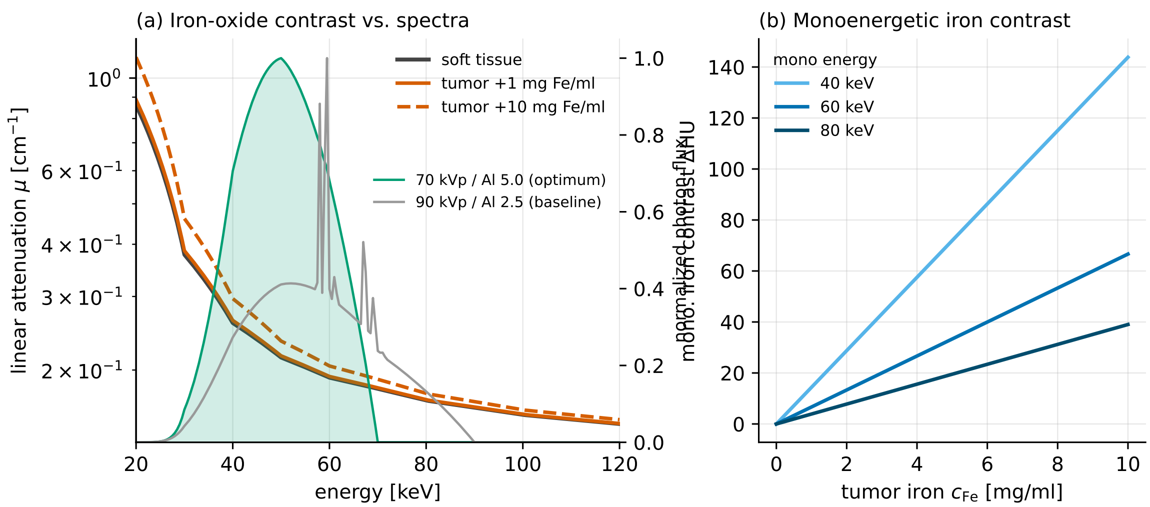

Figure 1. Physics of iron contrast (E1). Left: linear attenuation versus energy for soft tissue, cortical bone, and iron-loaded tumor at several concentrations. Iron has no usable K-edge (7.1 keV); the excess attenuation is photoelectric and concentrated at low energy. Right: per-energy iron contrast through the phantom, which motivates a soft spectrum and low-energy PCD bins.

2.2 Digital phantom

We use a rabbit-scale, electron-density-style calibration phantom. It is a 110 mm soft-tissue cylinder. It holds the iron-concentration inserts on a common ring, so beam-hardening cupping is common-mode across inserts. It also holds a cortical-bone rod as a deliberate beam-hardening source, which makes the bone factor meaningful. Each insert is a disk of iron-loaded soft tissue. Iron is added to the tissue matrix by the mixture rule, so a zero-iron insert equals background exactly. Soft tissue is approximated by a water-equivalent CONRAD body material, which keeps the zero-iron baseline clean. The tumor volume corresponds to an 8 cm^3 sphere ($r = 12.4$ mm). The reconstruction field of view is 20 cm. Figure 2 shows an axial slice and an example EID/PCD reconstruction.

2.3 CONRAD-native fan-beam pipeline

Every reconstruction step is a CONRAD class, in CONRAD's geometry convention. We do not hand-roll ramp filters, Parker weights, or geometry. The fan-beam geometry uses a source-to-isocentre distance of 750 mm and a

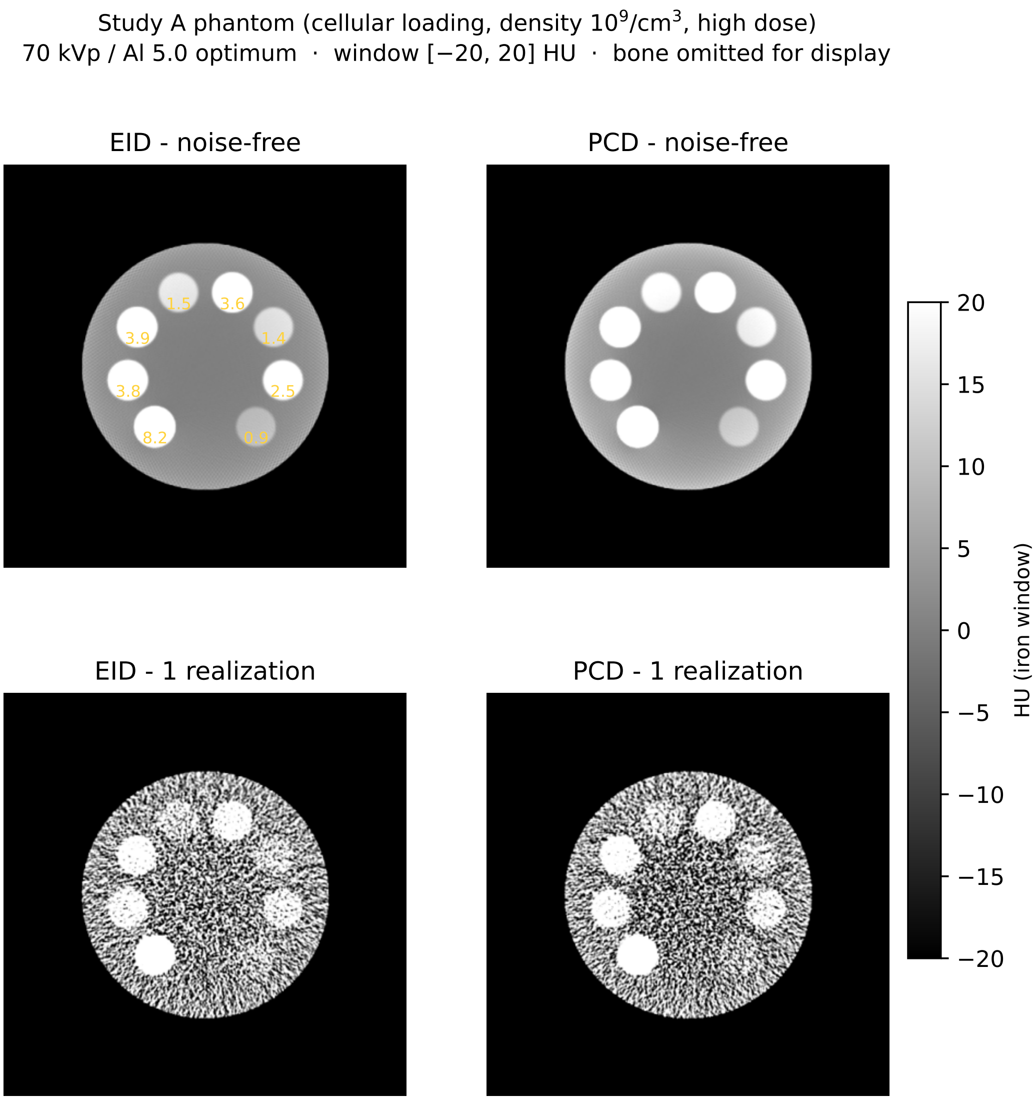

Figure 2. Digital phantom and CONRAD-native reconstruction. Left: axial slice of the rabbit-scale phantom – a soft-tissue cylinder with iron-concentration inserts on a ring and a cortical-bone rod. Center/right: example EID and PCD reconstructions of the same slice through the fan-beam short-scan pipeline (water precorrection on).

virtual detector at the isocentre sampled at $\Delta_T = 1\,\text{mm}$. We use 500 views over a minimal short scan (180° plus fan), with Parker redundancy weighting. The chain is: forward projection, Parker weighting, cosine filter, Shepp-Logan ramp filter, and a distance-weighted fan-beam backprojector. Reconstruction is on a 0.5 mm isotropic grid. A water beam-hardening precorrection is *always on*. The precorrection polynomial is calibrated so a water cylinder reconstructs flat to about 0.1%; it is not a factor of the study.

The GPU backprojector required a fix to the CONRAD OpenCL kernel, which never received the reconstruction pixel spacing; the corrected kernel matches the CPU reference to about 10^{-5} and is roughly 850× faster. It is contributed upstream, together with the analytic fan projector, the reinstated distance weighting, and the water precorrection tool.

2.4 Detectors: EID and PCD

Both detectors start from the same polychromatic forward model. For each energy E the transmitted photons are $N(E) = N_0\, s(E)\, e^{-\tau(E)}$, with s the normalized spectrum and τ the material path integral. Quantum noise is Poisson per energy.

The energy-integrating detector (EID) weights photons by energy: $S = \sum_E N(E)\,E$. Its variance is the exact second moment $\sum_E N(E)\,E^2$, not a single Poisson draw on the summed signal. The line integral is $p = -\log(S/S_{\mathrm{air}})$. A single water precorrection linearizes the response.

The photon-counting detector (PCD) sorts photons into three energy bins. We combine the bins with a *post-combination matched filter*. We first form a matched-filter weighted sum of the bin counts, then take a single logarithm, then apply one water precorrection on the combined spectrum. The matched-filter weights $w_b \propto S_b/V_b$ are the CNR-optimal weighting that reaches the ideal-observer ceiling. Combining in the count domain before the logarithm is robust to the photon-starved low-energy bin, whose per-bin logarithm would diverge when its counts reach zero. This post-combination scheme gives about 1.2× the EID CNR. We note that a per-bin pre-sum correction, where each bin is corrected on its own spectrum before the sum, is noisier and gives only about 1.0×; we therefore do not use it in the noisy study.

2.5 Experiments

We run five experiments. Table 2 lists the shared simulation parameters.

E1 – material and physics validation. We confirm the magnetite attenuation against NIST, confirm the iron K-edge is out of range, and confirm the whole-particle model keeps the attenuation iron-dominated (Fig. 1, Table 1).

E2 – spectrum and detector optimization. Iron contrast is photoelectric, so it favours low energy. We sweep the feasible C-arm tube voltage (70–120 kVp) crossed with inherent aluminium filtration (Al 1.0–8.0 mm) for both detectors, end-to-end through the pipeline. We drop hardening metals (Cu, Sn); they only hurt iron contrast. E2 then (a) picks the CNR-optimal spectrum, (b) re-optimizes the PCD three-bin thresholds for that spectrum, and (c) feeds the winning setting to E3, E4, and E5. The optimum is a soft 70 kVp beam with 5 mm aluminium. All later experiments run at this E2 optimum *and* at a 90 kVp baseline, so the optimum can be compared to a standard C-arm spectrum.

E3 – Study A, homogeneous cellular uptake. Iron internalized by tumor cells gives a roughly uniform tumor iron distribution. We sample the article's measured cellular loading (Heinen et al., Fig. 5, B16-F10 melanoma; 0 h / 24 h): SPION I-113 nm 8.23/3.78, SPION II-115 3.86/1.52, II-98 3.60/1.37, II-76 2.51/0.85 pg Fe/cell. Each configuration converts to a tumor iron concentration through a tumor cell density. Cell density is uncertain, so it is a factor with three levels (10^8, 3×10^8, 10^9 cells/cm^3), which gives a detectability band. Each configuration uses its formulation's coating. We also cross the bone and dose factors. Table 3 lists the loading-by-density grid.

E4 – Study B, vascular / fresh injection. Freshly injected SPIONs are still in the blood carrier inside the vessels, before cellular uptake. The contrast is confined to 150 μm vessels occupying about 10% of the tumor volume, at a tenfold local concentration. The vessel-local level is the independent variable (5–15 mg Fe/ml). The 0.5 mm CT voxel cannot resolve the vessels, so each insert is homogenized to its tumor-mean iron, which is what the reconstruction sees to first order. We then add the second-order effects that the homogeneous model omits: the beam-hardening nonlinearity from concentrating iron, and the structural noise from the binomial per-voxel vessel count.

E5 – three-dimensional cone-beam verification. We verify the result in three dimensions on real anatomy. We use the post-mortem RabbitCT rabbit volume[5] and its C-arm geometry, decoded from the benchmark's projection matrices (745 mm source-to-isocentre, 1196 mm source-to-detector, 496 views over 198°). We forward-project poly-energetically, insert a material-separated 8 cm^3 SPION tumor, and reconstruct with cone-beam FDK. E5 also runs at the E2 optimum and the 90 kVp baseline.

2.6 Metrics

We score detectability with three quantities. The iron contrast ΔHU is the mean insert value minus a local annular background, corrected by subtracting the zero-iron insert so only iron remains. The CNR is that contrast divided by the background noise measured over noise realizations. An insert is called detectable when its CNR clears the Rose criterion; we report both CNR ≥ 3 and ≥ 5. We use 30 noise realizations per cell.

Table 2. Shared simulation parameters. The E2 optimum and a 90 kVp baseline are both run in E3–E5.

Parameter	Value
Spectrum	E2 optimum (70 kVp, Al 5 mm) and 90 kVp baseline, W anode
Photons per pixel	low 20,000 / high 100,000 (Poisson), factor
Geometry (2D)	C-arm fan beam, 500 views, minimal short scan, Parker weighted
Source-to-isocentre / detector	750 mm / 1200 mm, $\Delta_T = 1$ mm
Reconstruction (2D)	fan FBP (cosine + Shepp-Logan + distance backprojector), 0.5 mm voxels
Beam hardening	water precorrection always on
Phantom	110 mm soft-tissue cylinder + cortical-bone rod (factor)
Tumor	8 cm^3 ($r = 12.4$ mm), iron-loaded
Detectors	EID; 3-bin PCD (post-combination matched filter)
3D (E5)	RabbitCT geometry (745/1196 mm, 496 views, 198°), cone-beam FDK
Noise realizations	30 per cell

3. RESULTS

3.1 E2 spectrum and detector optimum

Because iron contrast is photoelectric, it should favour a soft spectrum. The E2 sweep confirms this end-to-end. CNR falls as tube voltage rises and as aluminium hardens the beam. The CNR-optimal spectrum is a soft 70 kVp beam with 5 mm aluminium. At this optimum, evaluated at the representative 1 mg Fe/ml, the EID reaches a CNR of 9.6 and the PCD 16.9, a gain of 1.20×; the matched-filter PCD reaches 98% of the ideal-observer ceiling. The PCD three energy bins re-optimize to edges of 10.0/37.5/47.5/70.0 keV for this spectrum. The 70 kVp optimum lifts the EID CNR by about 24% over the 90 kVp baseline (7.8 → 9.6). Figure 3 shows the CNR surface over tube voltage and filter for both detectors, with the selected optimum marked. Table 5 reports the EID versus PCD CNR at the optimum.

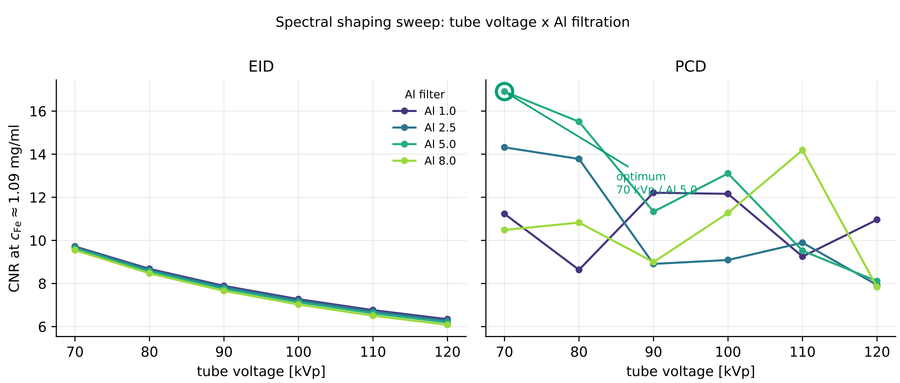

Figure 3. E2 spectral optimization. CNR versus tube voltage across aluminium filters, for EID and PCD. Lower tube voltage helps; hardening filters hurt. The selected CNR-optimal spectrum (70 kVp, Al 5 mm, PCD bin edges 10/37.5/47.5/70 keV) feeds E3–E5.

3.2 Study A: homogeneous cellular uptake and the cell-density band

The central result is that SPIONs sit near the CT detection limit at realistic load. Figure 4 plots iron ΔHU and CNR against tumor iron concentration for both detectors, with the Rose lines overlaid and a shaded cell-density band. The band comes from the three cell-density levels: the same cellular loading maps to different tumor iron concentrations as density changes, so visibility is dominated by how densely the tumor is packed with cells. The fresh SPION I loading (8.23 pg Fe/cell) gives tumor concentrations of 0.82/2.47/8.23 mg Fe/ml at the three densities. At high dose it clears the Rose CNR = 5 criterion already at 10^8 cells/cm^3, and at low dose only from

3×10^8 upward, so detection is dose-limited rather than detector-limited. The lowest detectable iron at Rose ≥ 5 is 0.41 mg Fe/ml for the PCD at the optimum (3×10^8, high dose) against 0.75 mg Fe/ml for the EID. At the reference load (0.54 mg Fe/ml) the iron contrast is only a few HU and the CNR sits on the Rose boundary; at the top of the density band the PCD CNR peaks near 96. Figure 5 shows a reconstruction montage across the cell-density levels. Table 3 lists the loading-by-density grid, and Table 4 the detection thresholds.

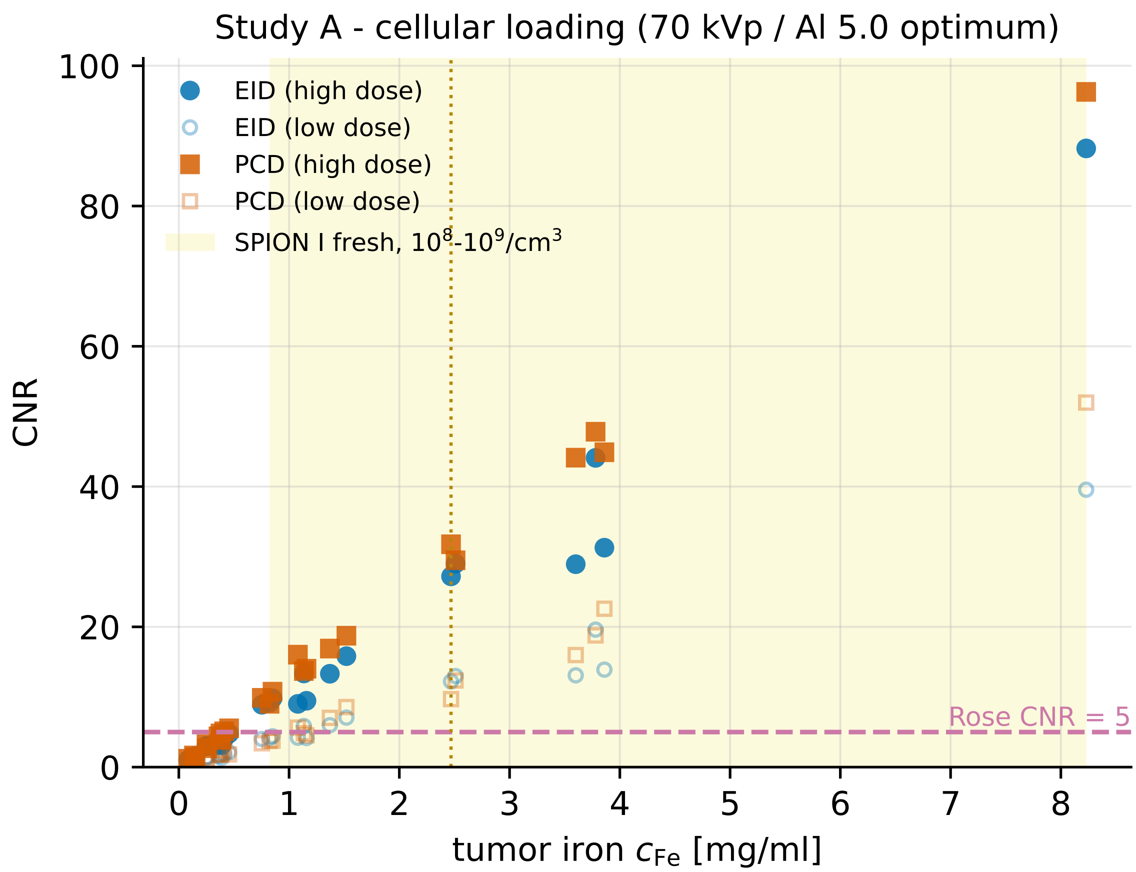

Figure 4. Study A detectability. Iron ΔHU (left) and CNR (right) versus in-tumor iron concentration for EID and PCD, with the Rose 3 and 5 lines. The shaded band spans the three tumor cell densities (10^8–10^9 cells/cm^3); it shows that visibility is dominated by cell density.

3.3 Study B: vascular / fresh injection

Real fresh-injection uptake is vascular, not uniform. The decisive fact is one of scale. A 0.5 mm voxel is more than three times wider than a 150 μm vessel, so each voxel partial-volume-averages many vessels. With mass conserved, the volume-averaged iron per voxel is unchanged, and the tumor-ROI contrast matches the homogeneous case. The two second-order effects are small: the beam-hardening nonlinearity is well below 10^{-3} HU, and the binomial per-voxel vessel count adds texture that averages out over the ROI. The ROI-mean detectability therefore tracks the total delivered iron, exactly as in Study A. At high dose the ROI-mean CNR crosses the Rose 5 line at a vessel-local level of 5 mg Fe/ml for the EID and 7 mg Fe/ml for the PCD; at low dose it crosses at 13 mg Fe/ml (EID) and 11 mg Fe/ml (PCD), so the PCD reaches threshold at a lower vessel level in the noisier low-dose regime. The PCD CNR peaks near 15.7 at the top vessel level and high dose. Figure 6 shows that vascular uptake is CT-indistinguishable from homogeneous uptake at realistic resolution.

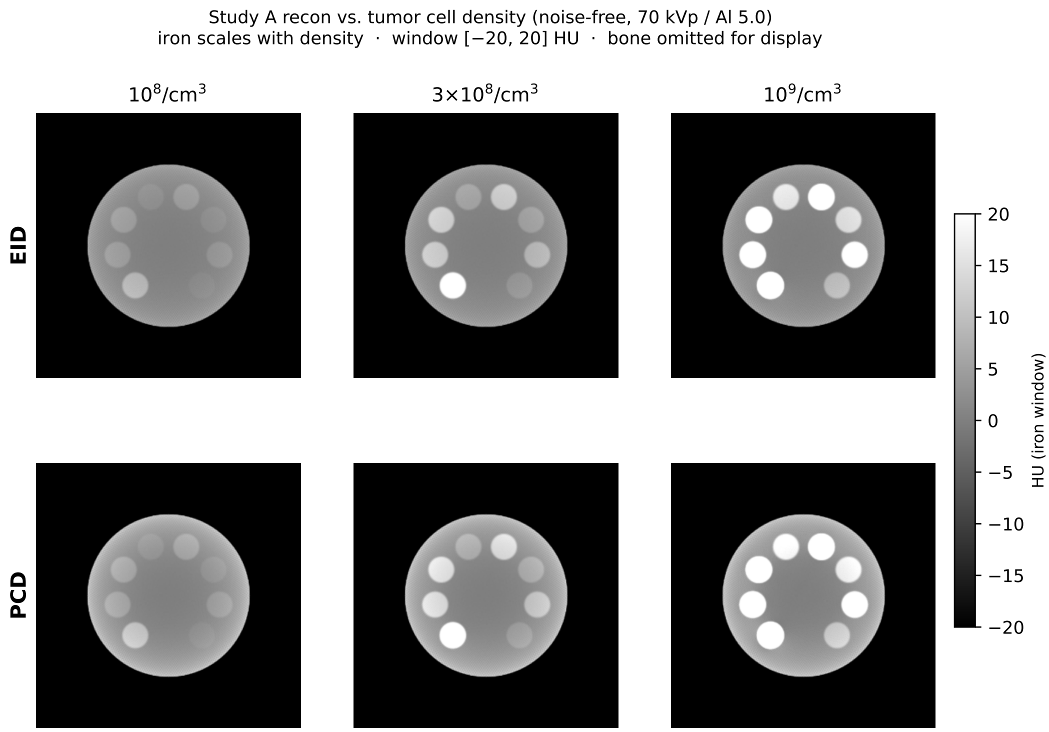

Figure 5. Study A cell-density montage. Reconstructions of the iron-loaded tumor at increasing tumor cell density ($10^8 \to 3 \times 10^8 \to 10^9$ cells/cm^3), at fixed cellular loading.

Table 3. Study A loading × cell-density grid. Measured cellular loading (Heinen et al., Fig. 5; 0 h / 24 h) converted to tumor iron concentration c_{Fe} [mg Fe/ml] via cell density ρ.

			c_{Fe} [mg/ml] at cell density		
Configuration	Formulation	pg Fe/cell	10^8	$3{\times}10^8$	10^9
I_113 (0 h)	I	8.23	0.82	2.47	8.23
I_113 (24 h)	I	3.78	0.38	1.13	3.78
II_115 (0 h)	II	3.86	0.39	1.16	3.86
II_115 (24 h)	II	1.52	0.15	0.46	1.52
II_98 (0 h)	II	3.60	0.36	1.08	3.60
II_98 (24 h)	II	1.37	0.14	0.41	1.37
II_76 (0 h)	II	2.51	0.25	0.75	2.51
II_76 (24 h)	II	0.85	0.09	0.26	0.85

3.4 EID versus PCD

Photon counting helps, but only partly. The post-combination matched-filter PCD gives about 1.2× the EID CNR when the low-energy bin carries enough counts. At the realistic dose the low-energy bin is photon-starved through the body, so the gain shrinks. Averaged over the Study A grid (bone off), the PCD CNR is about 1.15× the EID at the optimum and 1.28× at the baseline, a mean of about 1.2×; the softer optimum lifts the EID more, which narrows the relative gap. The per-bin pre-sum variant, which corrects each bin on its own spectrum before the sum, gives essentially no gain ($\sim 1.0\times$) and is noisier, so it is not used. Figure 7 shows where the iron contrast lives in energy and the re-optimized bin thresholds; the gain sits in a band that receives few transmitted photons. Table 5 reports the EID versus PCD CNR.

Table 4. Detection thresholds – the lowest c_{Fe} [mg Fe/ml] reaching the Rose CNR ≥ 5 criterion (bone off) – per detector, cell density, dose, and spectrum. A dash means the criterion is not reached within the swept range.

		Optimum (70 kVp/Al5)		Baseline (90 kVp/Al2.5)	
Detector	Cell density	low dose	high dose	low dose	high dose
EID	10^8	–	0.82	–	0.82
EID	3×10^8	1.13	0.75	2.47	0.75
EID	10^9	1.37	0.85	1.52	0.85
PCD	10^8	–	0.82	–	0.82
PCD	3×10^8	1.08	0.41	1.13	0.75
PCD	10^9	1.37	0.85	1.37	0.85

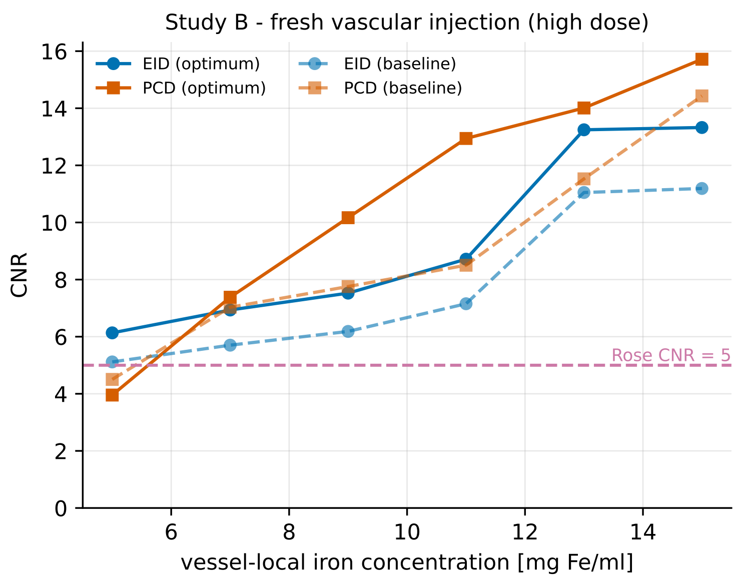

Figure 6. Study B, vascular uptake. The ROI-mean iron concentration is invariant (mass conservation), while the per-voxel peak approaches the resolved tenfold value only as the voxel shrinks toward the 150 μm vessel size. At the 0.5 mm CT voxel the vessels are unresolved.

3.5 Three-dimensional verification

Finally we verify the result in real anatomy. Casting rays through the RabbitCT volume with the decoded C-arm geometry reproduces the skull and cervical spine, and they rotate correctly with view angle, validating the geometry end to end. Inserting an 8 cm^3 SPION tumor makes it project as a localized signal that tracks the anatomy. [**TODO: E5 3D results pending: report the poly-energetic FDK iron ΔHU and CNR at the E2 optimum and at the 90 kVp baseline, and confirm consistency with the 2D detection limit.**] Figure 8 shows the cone-beam projections; the reconstructed-tumor panel is pending.

4. DISCUSSION

The picture is coherent. At biologically realistic load, SPIONs deposit little iron. That iron produces only a few Hounsfield units of contrast. The resulting CNR lands near the Rose criterion, not comfortably above it. So the reference dose is borderline-visible: detectable in principle at low noise and a soft spectrum, but not a robust everyday finding. The top iron concentration we study is still about an order of magnitude below iodinated-contrast enhancement, so marginal detectability is expected.

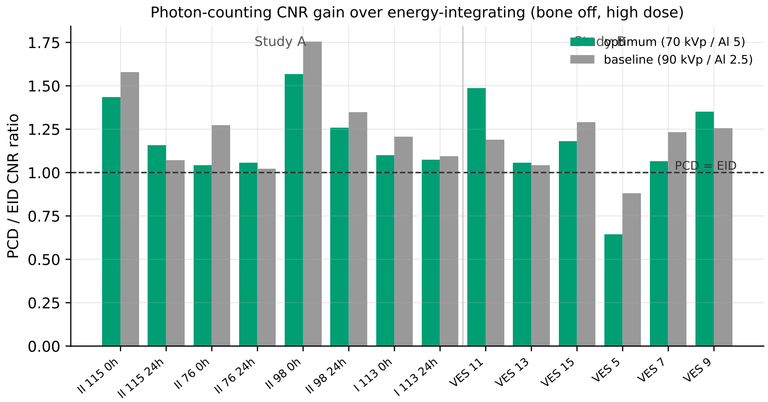

Figure 7. EID versus PCD. Per-energy iron detectability through the phantom, with the re-optimized three-bin PCD thresholds (10/37.5/47.5/70 keV). The contrast lives where transmitted photons are scarce, which limits the realized PCD gain to about 1.2× EID.

Table 5. EID versus PCD CNR by cellular-loading configuration, at cell density 10^9 cells/cm^3, high dose, bone off, for the E2 optimum and the 90 kVp baseline. The PCD uses the post-combination matched filter.

	Optimum (70 kVp/Al5)			Baseline (90 kVp/Al2.5)	
Configuration	EID	PCD	PCD/EID	EID	PCD
I 113 0h	88.2	96.3	1.09	72.1	92.6
I 113 24h	44.1	47.8	1.08	35.8	38.5
II 115 0h	31.3	44.9	1.43	25.9	42.5
II 115 24h	15.8	18.7	1.18	13.0	13.1
II 98 0h	28.9	44.1	1.53	24.0	42.8
II 98 24h	13.3	16.9	1.27	11.0	14.5
II 76 0h	29.0	29.5	1.02	24.0	32.1
II 76 24h	9.8	10.8	1.10	8.1	8.4

Which factor helps most? Cell density. In Study A the same cellular loading maps to a wide range of tumor iron concentrations depending on how densely the tumor is packed. Visibility is therefore dominated by tumor cell density, not by the formulation alone. The next lever is the spectrum. A soft beam (low tube voltage, no hardening filtration) raises iron contrast because the contrast is photoelectric. This is consistent across the E2 sweep, the ideal-observer analysis, and physical intuition. The photon-counting detector gives a modest gain of about 1.2×, and only when the low-energy bin survives the body; the noisier per-bin correction gives essentially nothing. Beam-hardening correction does not change iron detectability, because iron contrast is a genuine material signal, not a cupping artifact.

We state the limitations candidly. The study is a simulation with idealized, ideal-energy detectors and no scatter or patient motion. The 2D phantom is a round, geometric, rabbit-scale cylinder rather than full rabbit anatomy, so beam hardening and scatter are idealized. Streaks from the bone rod are suppressed but not eliminated by the local-background and zero-iron subtraction. Soft tissue is a water-equivalent proxy. The homogeneous-uptake assumption is relaxed by Study B, which shows that sub-resolution vessels do not change detectability at the same total dose. The E5 cone-beam verification brings in real anatomy, but a full 3D detectability sweep is still future work. The open decisions are the tumor cell density used in Study A and the injection concentration used in Study B; both are reported as bands rather than single numbers.

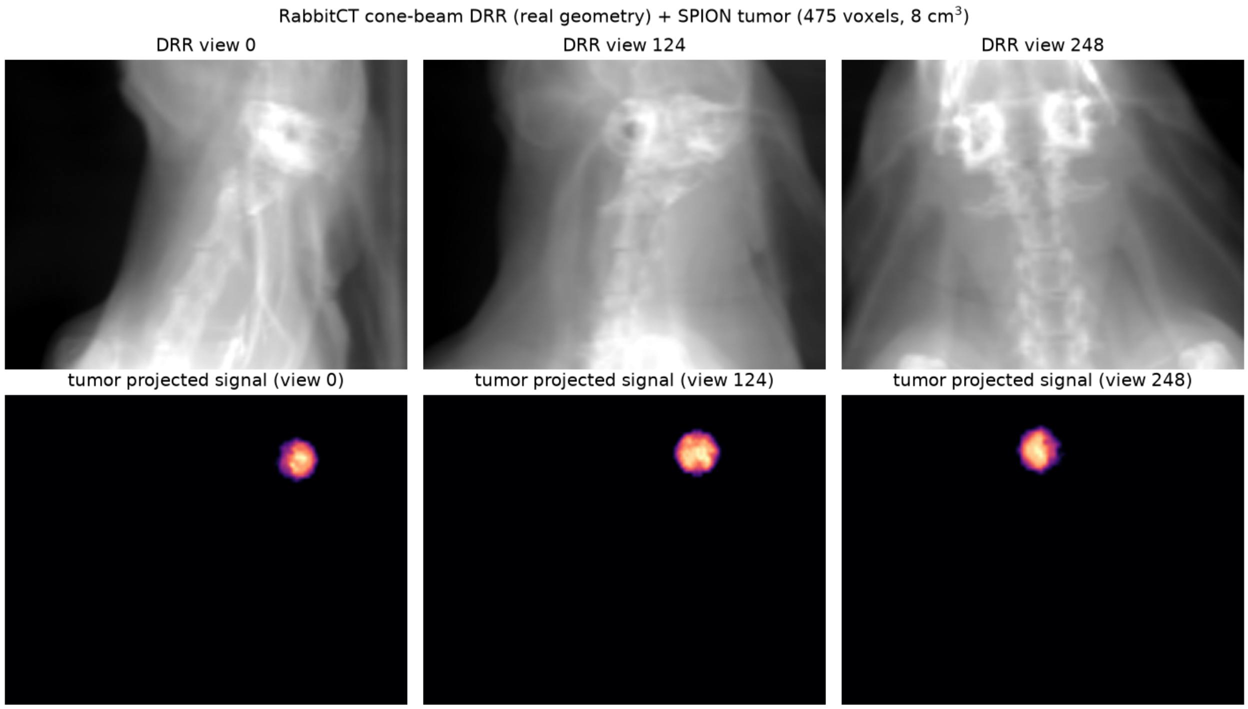

Figure 8. E5 three-dimensional verification. Top: ray-driven cone-beam projections of the post-mortem RabbitCT rabbit at three view angles, using the decoded C-arm geometry (745/1196 mm, 496 views over 198°). Bottom: the projected signal of an inserted 8 cm^3 SPION tumor, which tracks the anatomy. [**TODO: add the poly-energetic FDK reconstruction with the iron ROI**]

5. CONCLUSION

Can superparamagnetic iron-oxide nanoparticles be seen in CT? At realistic biological load, only barely. Our CONRAD-native simulation places the detection limit near the reference SPION dose, with an iron contrast of a few Hounsfield units and a CNR on the Rose boundary. Visibility is dominated by tumor cell density. Photon counting adds about 1.2×; a soft spectrum is the most effective lever; beam-hardening correction does not move the limit. For image-guided nanoparticle therapy this means routine CT confirmation of SPION delivery is feasible only at the upper end of realistic loading and with a spectrum tuned for iron. The path forward is a full three-dimensional anatomical detectability study and experimental validation against phantom and *ex vivo* measurements. All of the simulation machinery used here, including GPU-native spectral detectors and noise generators contributed to CONRAD, is openly available to support that work.

Code and data availability

The complete simulation pipeline, intermediate results, and figure sources are public, with an audit dashboard at https://akmaier.github.io/SPIONvsXRay/. The project is MIT-licensed. The GPU OpenCL detector, noise generator, analytic fan projector, water precorrection tool, and back-projector fixes have been upstreamed to CONRAD.

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